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CHEMISTRY

Chemistry: Matter and Change — Part 3

Class 11

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Chapter 1

Some Basic Concepts of Chemistry

Chemistry deals with the composition, structure, properties and transformation of matter. Matter is classified into elements, compounds and mixtures. The mole concept relates the mass of a substance to the number of particles it contains, with Avogadro's number (6.022 x 10^23) bridging the atomic and macroscopic scales. Chemical equations represent reactions quantitatively, and stoichiometry allows calculation of reactants and products. Understanding these concepts is fundamental to all branches of chemistry and to industrial applications.

Chapter 2

Structure of Atom

The atom consists of a dense nucleus of protons and neutrons, surrounded by electrons. Dalton's atomic theory, Thomson's plum pudding model, Rutherford's nuclear model and Bohr's planetary model successively refined our understanding. Quantum mechanics describes electrons as occupying orbitals defined by probability distributions. The arrangement of electrons in shells and subshells determines chemical behavior. The modern periodic table organizes elements by atomic number, revealing periodic trends in atomic radius, ionization energy and electronegativity.

Chapter 3

Chemical Bonding

Atoms combine to achieve stable electron configurations, forming chemical bonds. Ionic bonds result from the transfer of electrons between metals and non-metals, as in sodium chloride. Covalent bonds involve the sharing of electrons, as in water and methane. Metallic bonds account for the properties of metals. The shape of molecules, predicted by VSEPR theory, influences polarity and reactivity. Understanding bonding explains the properties of substances and underlies all chemical reactions.